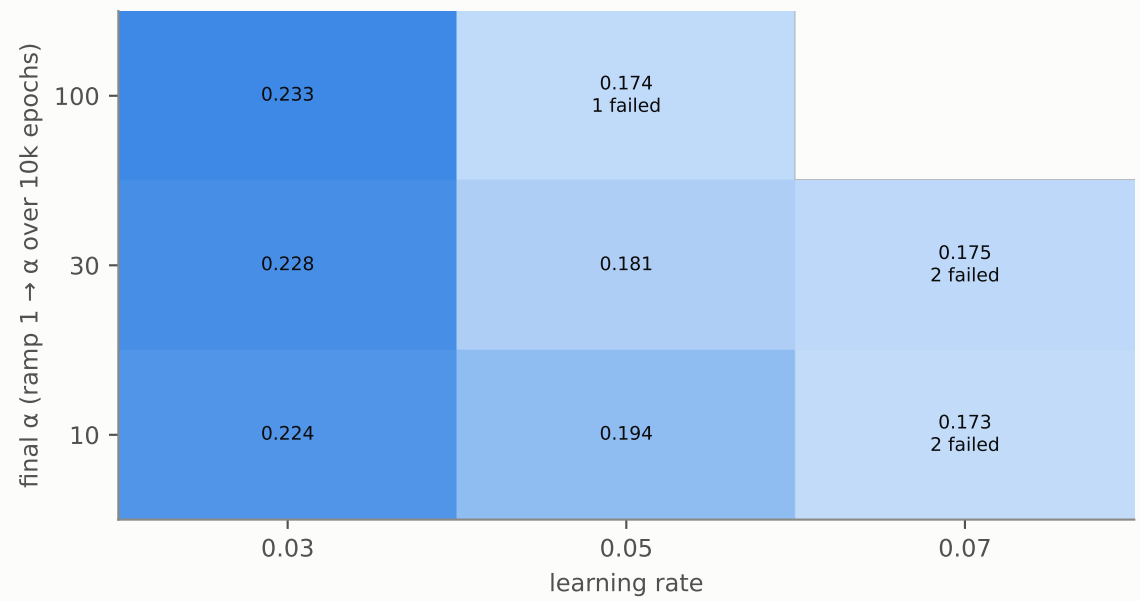
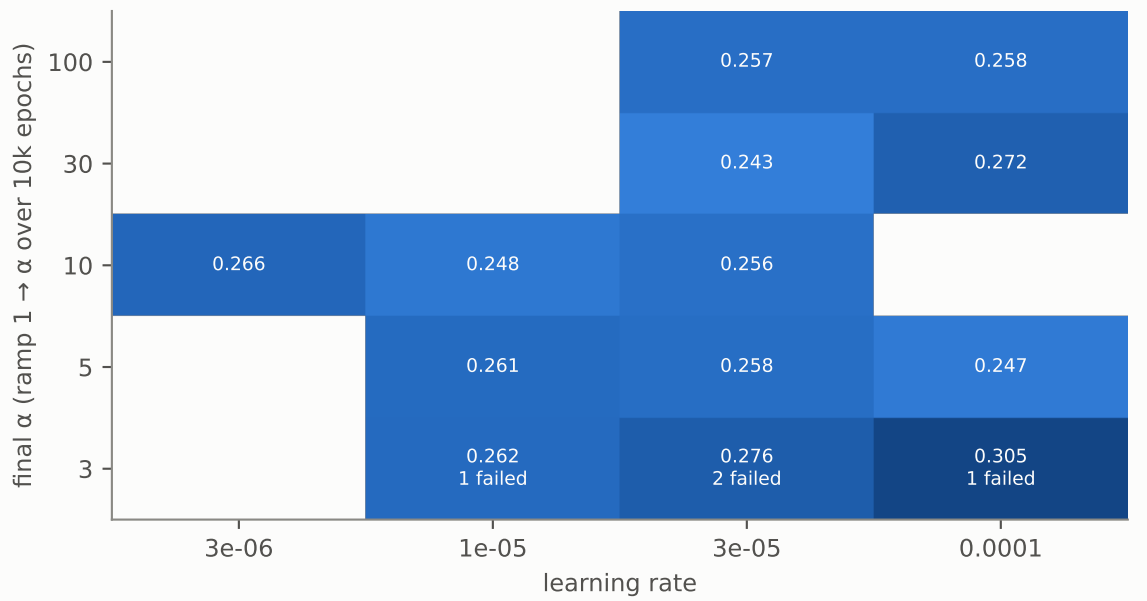
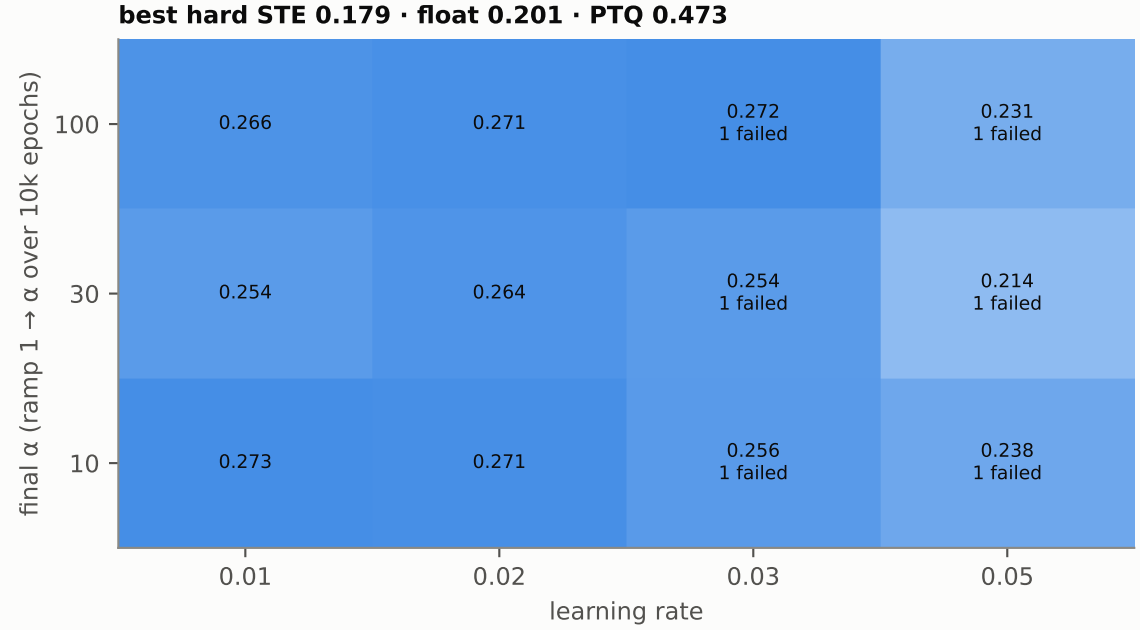
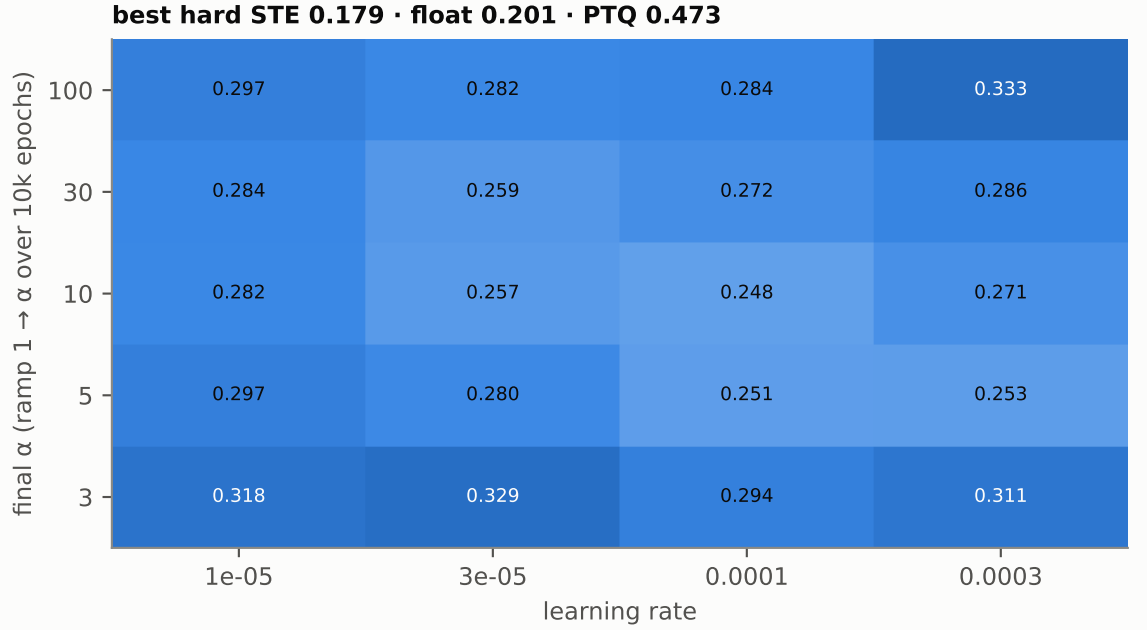
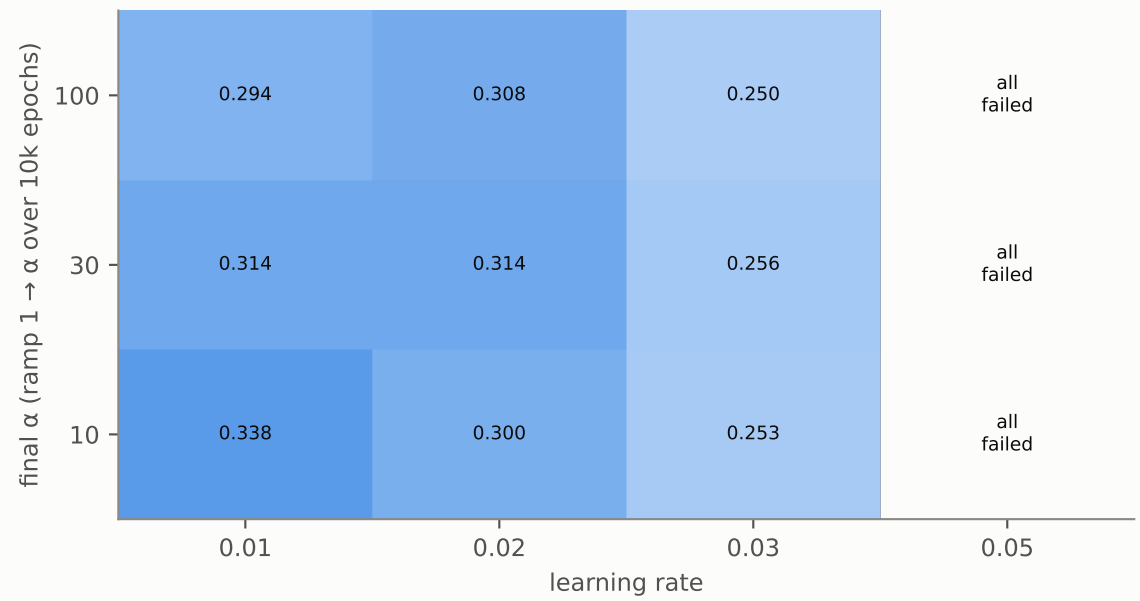
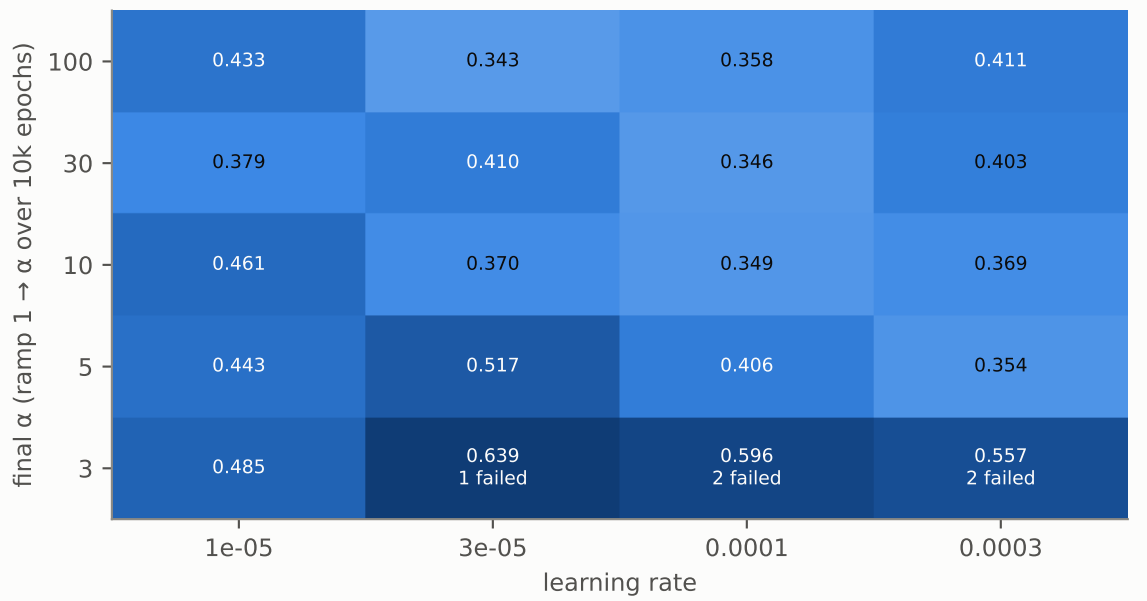

8/4 bits (total/frac) · slope 1 (soft STE)
best hard STE 0.177 · float 0.201 · PTQ 0.327
final α (ramp 1 → α over 10k epochs)
100
30
10
0.233
0.174
1 failed
0.228
0.181
0.175
2 failed
0.224
0.194
0.173
2 failed
0.03
0.05
0.07
learning rate

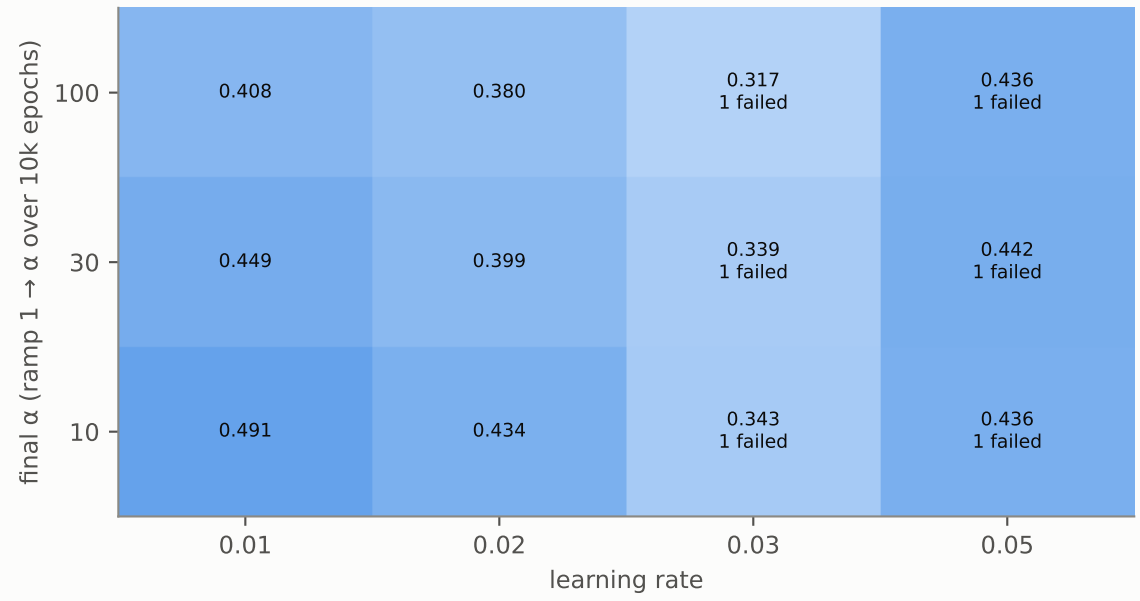

8/4 bits (total/frac) · exact, β = 1
best hard STE 0.177 · float 0.201 · PTQ 0.327
final α (ramp 1 → α over 10k epochs)
100
30
10
5
3
0.257
0.258
0.243
0.272
0.266
0.248
0.256
0.261
0.258
0.247
0.262
1 failed
0.276
2 failed
0.305
1 failed
3e-06
1e-05
3e-05
0.0001
learning rate

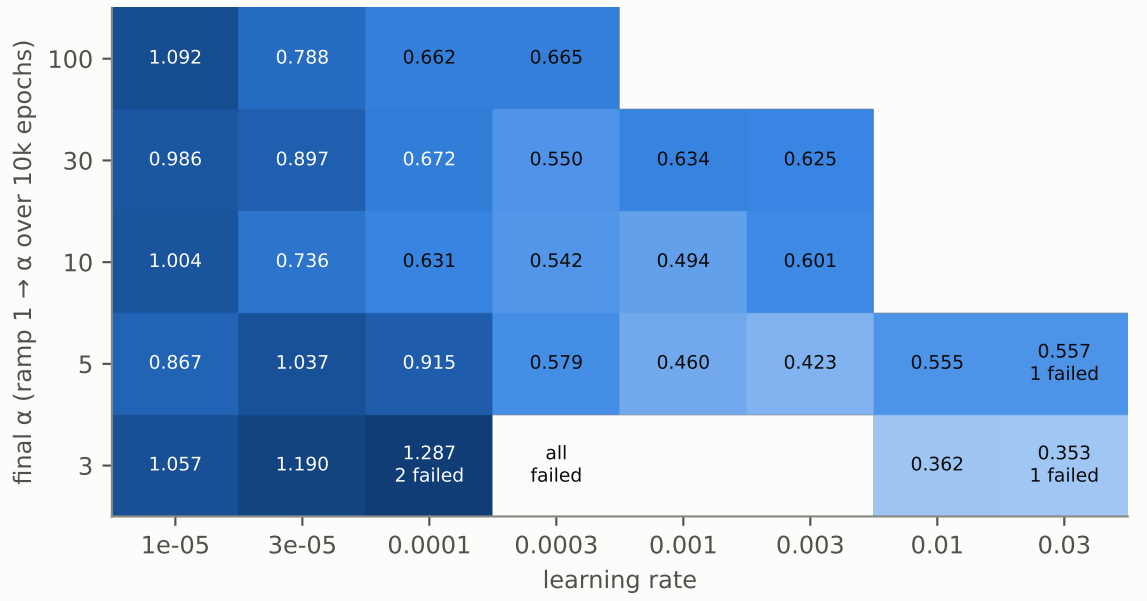

6/4 bits (total/frac) · slope 1 (soft STE)
best hard STE 0.179 · float 0.201 · PTQ 0.473
final α (ramp 1 → α over 10k epochs)
100
30
10
0.266
0.271
0.272
1 failed
0.231
1 failed
0.254
0.264
0.254
1 failed
0.214
1 failed
0.273
0.271
0.256
1 failed
0.238
1 failed
0.01
0.02
0.03
0.05
learning rate


6/4 bits (total/frac) · exact, β = 1
best hard STE 0.179 · float 0.201 · PTQ 0.473
final α (ramp 1 → α over 10k epochs)
100
30
10
5
3
0.297
0.282
0.284
0.333
0.284
0.259
0.272
0.286
0.282
0.257
0.248
0.271
0.297
0.280
0.251
0.253
0.318
0.329
0.294
0.311
1e-05
3e-05
0.0001
0.0003
learning rate


6/3 bits (total/frac) · slope 1 (soft STE)
best hard STE 0.229 · float 0.201 · PTQ 0.672
final α (ramp 1 → α over 10k epochs)
100
30
10
0.294
0.308
0.250
all
failed
0.314
0.314
0.256
all
failed
0.338
0.300
0.253
all
failed
0.01
0.02
0.03
0.05
learning rate


6/3 bits (total/frac) · exact, β = 1
best hard STE 0.229 · float 0.201 · PTQ 0.672
final α (ramp 1 → α over 10k epochs)
100
30
10
5
3
0.433
0.343
0.358
0.411
0.379
0.410
0.346
0.403
0.461
0.370
0.349
0.369
0.443
0.517
0.406
0.354
0.485
0.639
1 failed
0.596
2 failed
0.557
2 failed
1e-05
3e-05
0.0001
0.0003
learning rate


4/2 bits (total/frac) · slope 1 (soft STE)
best hard STE 0.288 · float 0.201 · PTQ 1.393
final α (ramp 1 → α over 10k epochs)
100
30
10
0.408
0.380
0.317
1 failed
0.436
1 failed
0.449
0.399
0.339
1 failed
0.442
1 failed
0.491
0.434
0.343
1 failed
0.436
1 failed
0.01
0.02
0.03
0.05
learning rate


4/2 bits (total/frac) · exact, β = 1
best hard STE 0.288 · float 0.201 · PTQ 1.393
final α (ramp 1 → α over 10k epochs)
100
30
10
5
3
1.092
0.788
0.662
0.665
0.986
0.897
0.672
0.550
0.634
0.625
1.004
0.736
0.631
0.542
0.494
0.601
0.867
1.037
0.915
0.579
0.460
0.423
0.555
0.557
1 failed
1.057
1.190
1.287
2 failed
all
failed
0.362
0.353
1 failed
1e-05
3e-05
0.0001
0.0003
0.001
0.003
0.01
0.03
learning rate